EXPERIMENT NO: 11

Experiment Title: Use JavaScript timers (setTimeout(), setInterval()) and ES6 modules (import/export) to build a dynamic countdown or class scheduler with real-time updates and responsive layout.

Software/Tools Required:

- Any text editor / VS Code
- Web Browser (Chrome/Edge) with Developer Console

Theory:

JavaScript Timers: JavaScript timers allow functions to execute after a specified delay or repeatedly at regular intervals. They are useful for implementing countdown timers, digital clocks, reminders, and class scheduling applications.

Key Timer Functions:

1. **setTimeout(): Executes a function once after a specified time delay. In this experiment, it is used to schedule a callback when the class is expected to begin.**
2. **setInterval(): Executes a function repeatedly at a specified interval. It is used to update the current clock and countdown display every second.**

3. **clearTimeout(): Cancels a previously scheduled timeout.**
4. **clearInterval(): Stops a repeating interval and prevents further executions.**

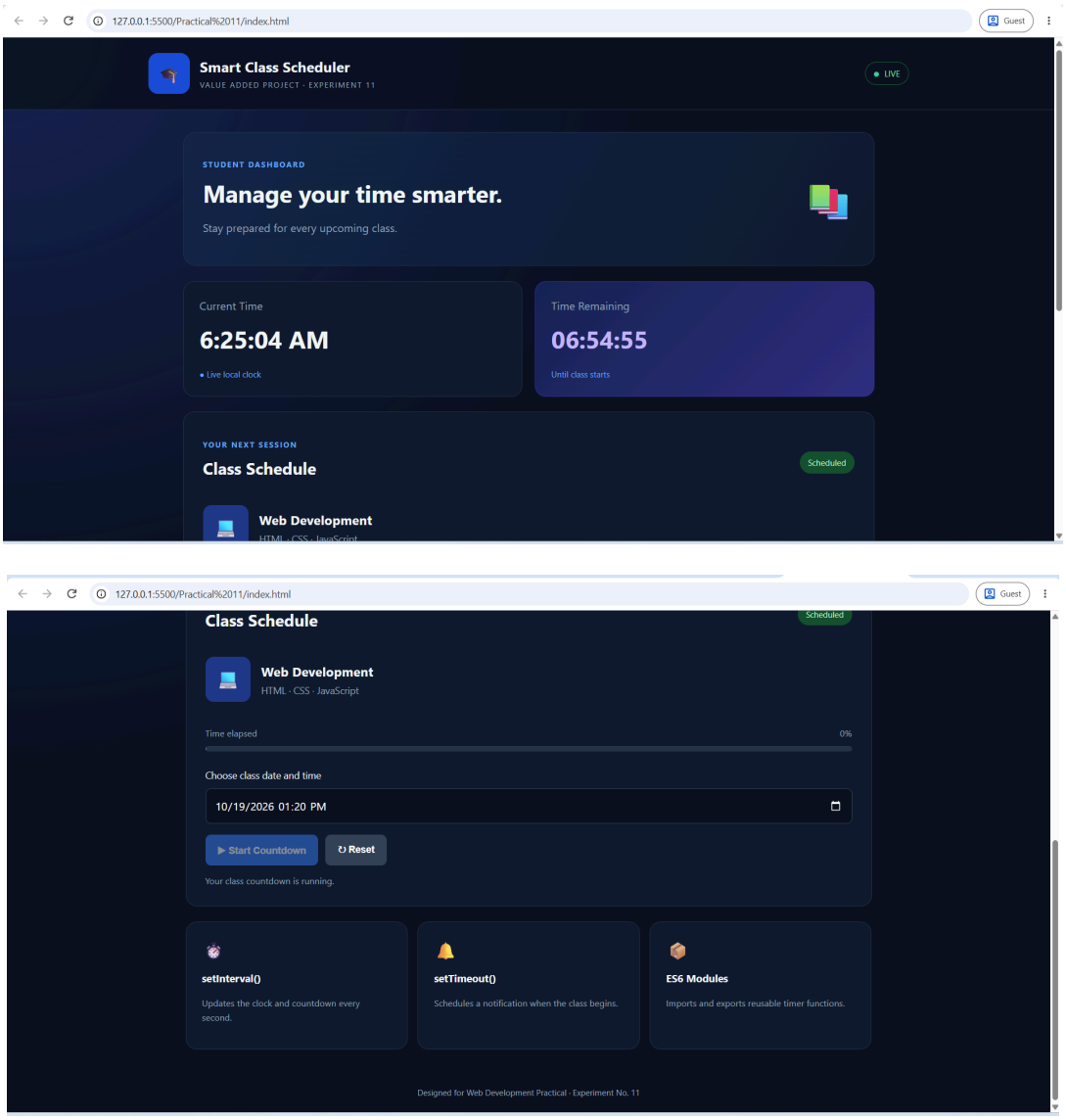
ES6 Modules (Import/Export): ES6 modules allow JavaScript code to be organized into separate files for better reusability and maintainability. The export keyword makes functions or variables available to other modules, while the import keyword allows them to be used in another JavaScript file. In this experiment, a reusable time-formatting function is exported from timer.js and imported into scheduler.js.

Dynamic Countdown and Class Scheduling: A countdown timer calculates the difference between the current time and the selected class start time. JavaScript updates the remaining time dynamically and displays the class status. When the countdown reaches zero, the

application displays a class-start notification.

Responsive Web Design: CSS media queries, flexible layouts, and responsive dimensions allow the class scheduler interface to adapt to different screen sizes, including desktops, tablets, and mobile devices.

OUTPUT :



RESULT :

Thus, a responsive Smart Class Scheduler was successfully implemented using JavaScript timers (setTimeout() and setInterval()) and ES6 modules (import/export) to provide real-time clock updates, countdown functionality, class scheduling, and dynamic status notifications.